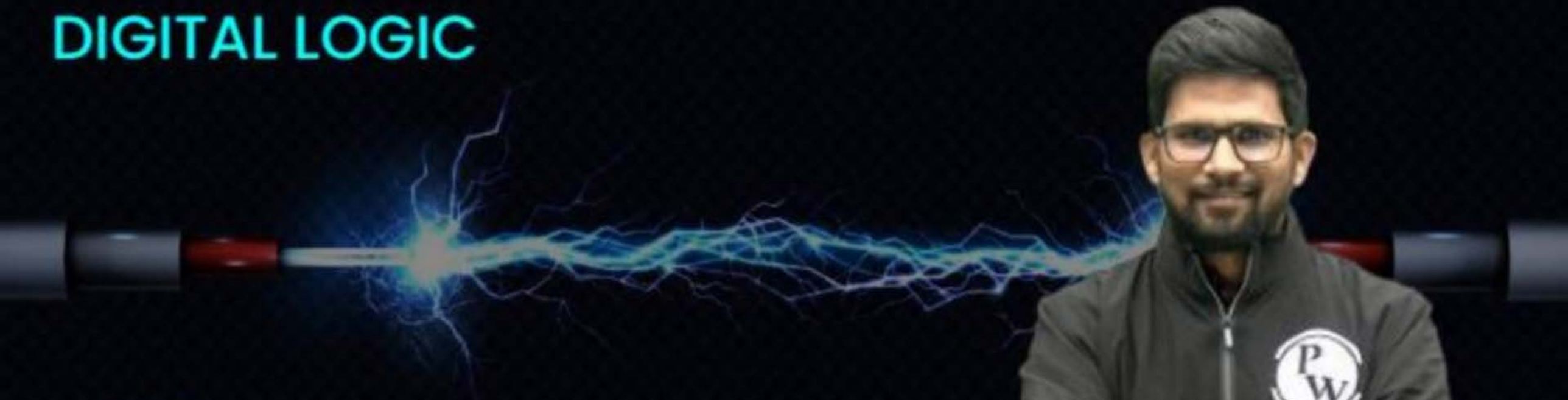
COMPUTER SCIENCE & IT

DIGITAL LOGIC



Lecture No: 07

Miscellaneous Topics



By- Chandan Gupta Sir

Recap of Previous Lecture

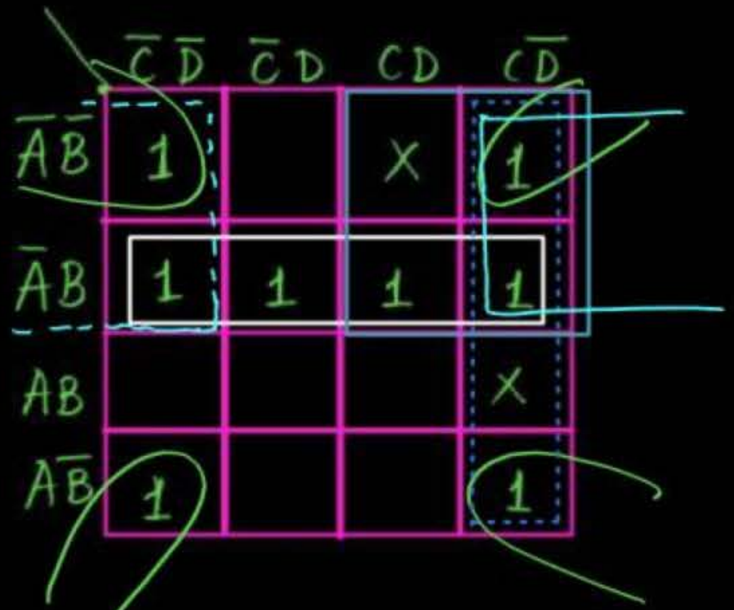


- State transition Diagram
- Parity generator & checker
- T_0, T_1, L, M



Functionally complete function

#Q.

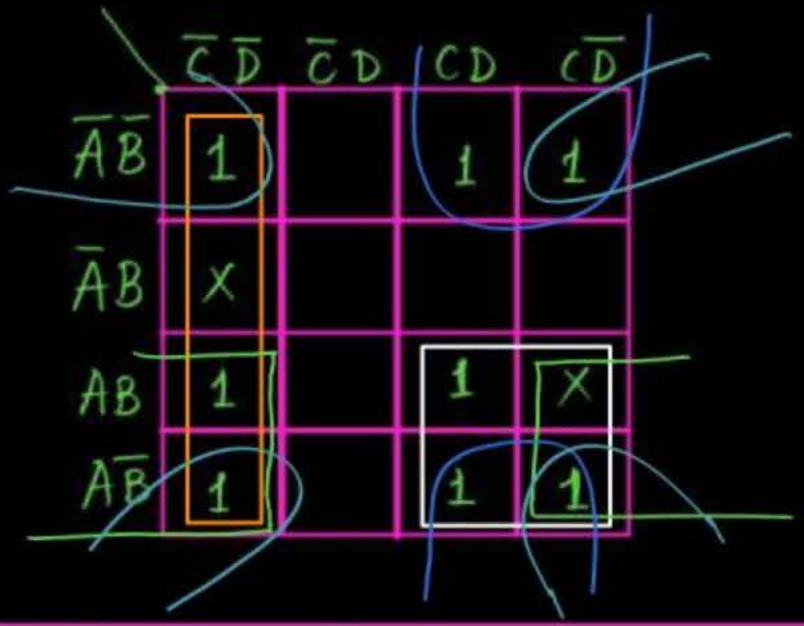


$PI_s \rightarrow \bar{A}B, \bar{B}\bar{D}, C\bar{D}, \bar{A}C, \bar{A}\bar{D}$

$EPI_s \rightarrow \bar{A}B, \bar{B}\bar{D}$

H.W.

#Q.

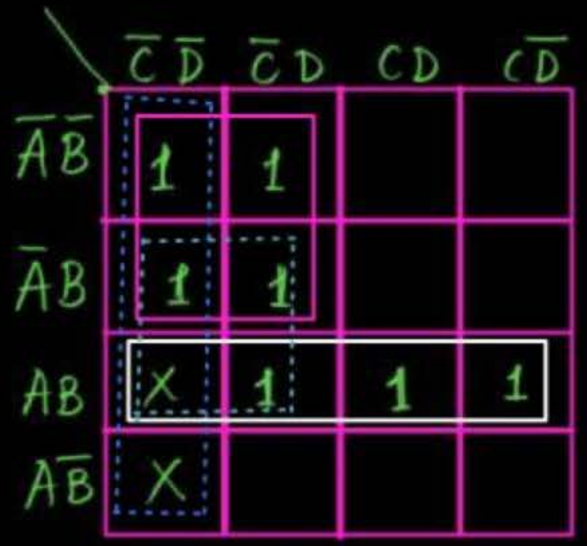


$PI_s \longrightarrow AC,\ \bar{B}C,\ \bar{C}\bar{D},\ A\bar{D},\ \bar{B}\bar{D}$

$EPI_s \longrightarrow AC,\ \bar{B}C$

HW

#Q.



$PI_s \longrightarrow AB, \bar{A}\bar{C}, B\bar{C}, \bar{C}\bar{D}$

$EPI_s \longrightarrow AB, \bar{A}\bar{C}$

HW

H.W

Q.1 $f(A,B,C) = \overline{A} + \overline{B}C$, T_0, T_1, L, M

Not T_0 , Not T_1 , Not L , Not M

Q.2 $f(A,\overline{B}) = \overline{A}B$, T_0, T_1, L, M

Not T_0 , Not T_1 , Not L , Not M

Q.3 $f(\overline{A},B) = A + \overline{B}$, T_0, T_1, L, M

Not T_0 , Not T_1 , Not L , Not M

Q.4 $f(\overline{A},\overline{B}) = \overline{A}B$, T_0, T_1, L, M

T_0 , Not T_1 , Not L , Not M

Q.5 $f(A,\overline{B},C) = \overline{A}B + \overline{C}$, T_0, T_1, L, M

Not T_0 , Not T_1 , Not L , Not M

$f(A,\overline{B}) = \overline{A} + \overline{B}$

$\overline{B} = D$

$f(A,D) = \overline{A} + D$

$= \Sigma(0,1,3)$

Not T_0 , T_1 , NL , NM

00	—	1
01	→	1
10	—	0
11	—	1

Self Dual function:

- $f^D = f \longrightarrow$ self dual $\quad S_D$

  $f^D \neq f \longrightarrow$ Non - self dual (Not S_D)

Functionally Complete function :

- A boolean function is said to be functionally complete if it does not belong to any of function type T_0, T_1, L, M & S_D.
  - → If it belongs to any one of them then it will not be functionally complete function.

- $f = \{ f_1, f_2, f_3 \cdots \}$ → A set of boolean function is said to functionally complete if
  1. Atleast one function does not belong to T_0.
  2. Atleast one function does not belong to T_1.
  3. Atleast one function does not belong to L.
  4. Atleast one function does not belong to M.
  5. Atleast one function does not belong to S_D.

#Q. $f(A,B,C) = A + \overline{B}\,\overline{C} = \Sigma(0,4,5,6,7)$

Not T_0 Not L

T_1 Not S_D

N M

$f(0,0,0) = 1$

$f(1,1,1) = 1$

A

1 00 4

1 01 5

1 10 6

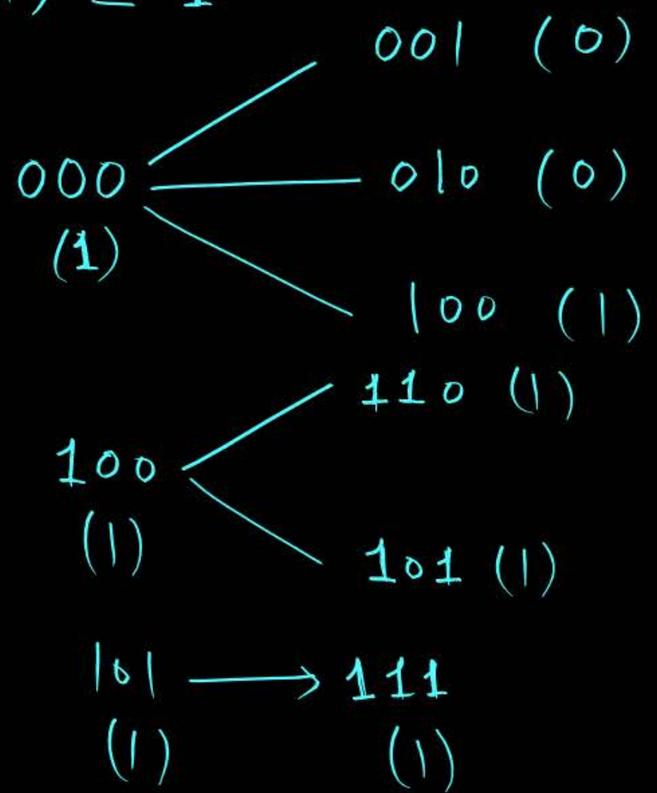
1 11 7

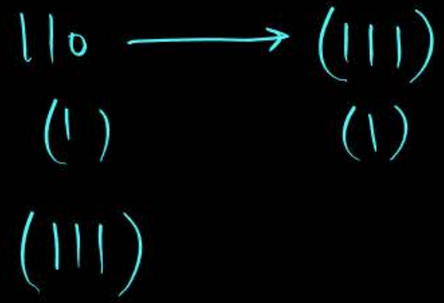
$\overline{B}\,\overline{C}$

0 00 → 0

1 00 → 4

Not functionally Complete function.





#Q. $f(A, B, \overline{C}) = \overline{A} + \overline{B}C$

$f(A, B, D) = \overline{A} + \overline{B}\,\overline{D} = \Sigma(0,1,2,3,4)$

$\overline{A}$

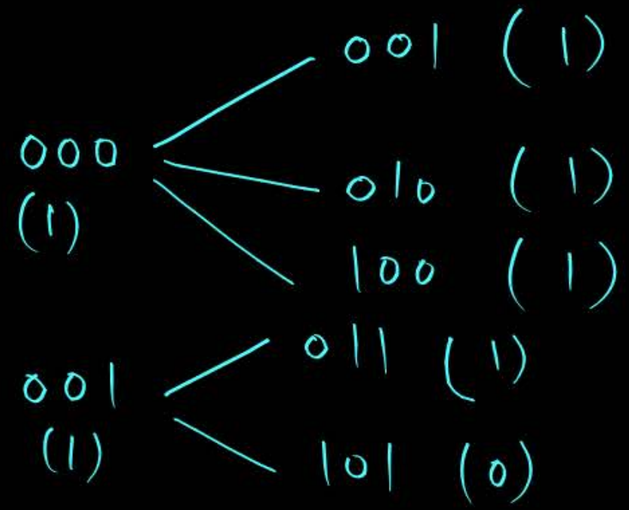
0 00 → 0
0 01 → 1
0 10 → 2
0 11 → 3

$\overline{B}\,\overline{D}$

0 0 0 → 0
1 0 0 → 4

$f(0,0,0) = 1$

$f(1,1,1) = 0$



Functionally Complete function.

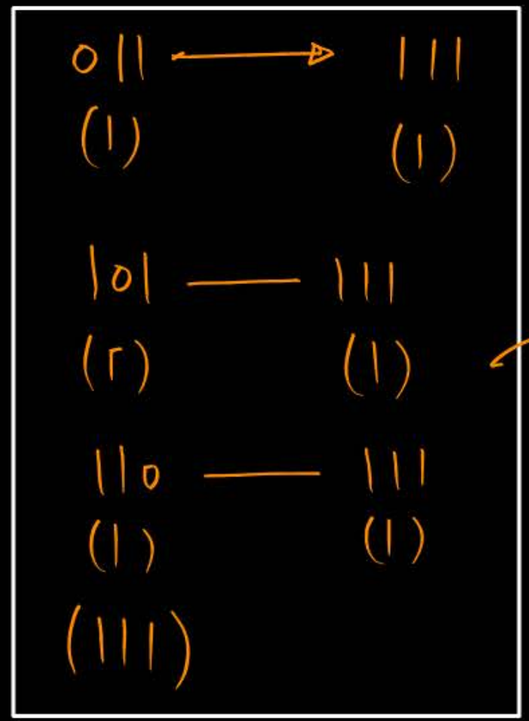
Not T_0
Not T_1
Not L
Not S_D
Not M

#Q. $f(A,B,C) = AB+BC+CA = \Sigma(3,5,6,7)$

3 — 011
5 — 101
6 — 110
7 — 111

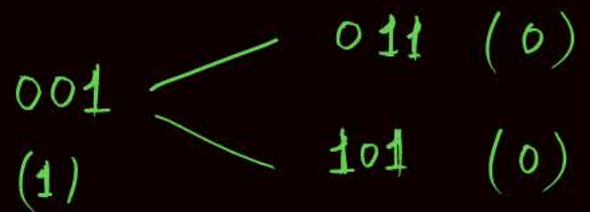
T_0
T_1
S_D
Not L
M

functionally incomplete function



Monotonic

Q. $f(A,B,C) = A \oplus B \oplus C = \Sigma(1,2,4,7)$



T_0

T_1

S_D

L

Not M.

functionally incomplete function.

#Q. $f(A,B) = (\bar{A}+B) = \Sigma(0,1,3)$

$\bar{A}$

0 0 $\longrightarrow$ 0

0 1 $\longrightarrow$ 1

B

0 1 $\longrightarrow$ 1

1 1 $\longrightarrow$ 3

(0 0) (1)

01 (1)

10 (0)

Not T_0

T_1

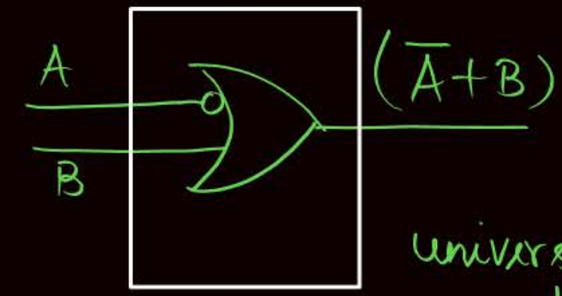
Not L

Not S_D

Not M

functionally incomplete function.

(A) $f(A,A) = \bar{A}+A = 1$



universal gate.

by default 0, 1 are available

- $f(\underline{A}, \underline{\bar{B}}) = \bar{A}B$

$f(A, C) = \bar{A}\bar{C} = \Sigma(0)$

00 (1) — 01 (0), 10 (0)

Not To
NoT T_1
Not S_D
Not L
Not M

functionally complete function.

A → $f(A, A) = \bar{A} \cdot \bar{A} = \bar{A}$

A, B $f(A, A) = \bar{A} = f_1$

$f(B, B) = \bar{B} = f_2$

$f(f_1, f_2) = \bar{f_1} \cdot \bar{f_2}$

$= A \cdot B$

A, B $f(A, B) = \bar{A} \cdot \bar{B} = f_3$

$f(f_3, f_3) = \bar{f_3} \cdot \bar{f_3} = \bar{f_3}$

$= \overline{\bar{A} \cdot \bar{B}} = (A + B)$

#Q. $f(A,B) = \bar{A}B + A\bar{B} = \Sigma(1,2)$

T_0

Not T_1

L

Not M

Not S_D.



functionally incomplete function.

Q. $f = \{ f_1(A,B) = A \oplus B, \quad f_2(A,B) = (A+B) \}$

T_0

Not T_1

L

Not S_D

Not M

$f_2(A,B) = A+B$

$= \Sigma(1,2,3)$

T_0

T_1

Not S_D

Not L

M.

Overall → functionally incomplete function.

$\# f = \{ f_1(A,B) = A \oplus B \;,\; f_2(A,B) = (A+B) \;,\; f_3(A) = \overline{A} \}$

$f_3(A) = \overline{A} = \Sigma(0)$

f_1	f_2	f_3
T_0	T_0	Not T_0
Not T_1	T_1	Not T_1
Not S_D	Not S_D	Not M
L	Not L	L
Not M	M	S_D

f_3
Not T_0
Not T_1
Not M
L
S_D

functionally complete function.

0 ⟶ 1

(1) 0

Implication function:

$P \longrightarrow Q$

P implies Q

if P then Q

P	Q	$f(P \to Q)$ [$f(P, Q)$]
0	0	1
0	1	1
1	0	0
1	1	1

$f(P, Q) = \Sigma(0, 1, 3) = \pi(2) = (\overline{P} + Q)$

↓ implication function

Not To

T_1

Not L

Not SD

Not M.

Negation function:

$$f(A) = \overline{A}$$

Not T_0

Not T_1

Not M.

L

S_D

[MCQ]

A set of Boolean connectives is functionally complete if all Boolean functions can be synthesized using those. Which of the following sets of connectives is NOT functionally complete?

A. EX-NOR → Not functionally complete

B. implication, Negation → functionally complete

C. OR, negation → functionally Complete

D. NAND → functionally complete

$f(A,B) = \overline{A \cdot B} = \Sigma(0,1,2)$

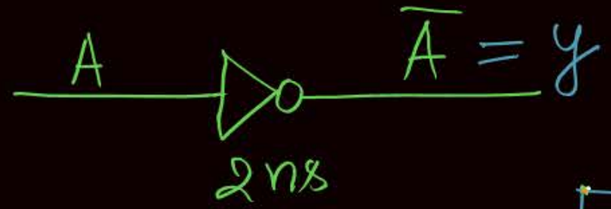
00 < 01 (1), 10 (1)

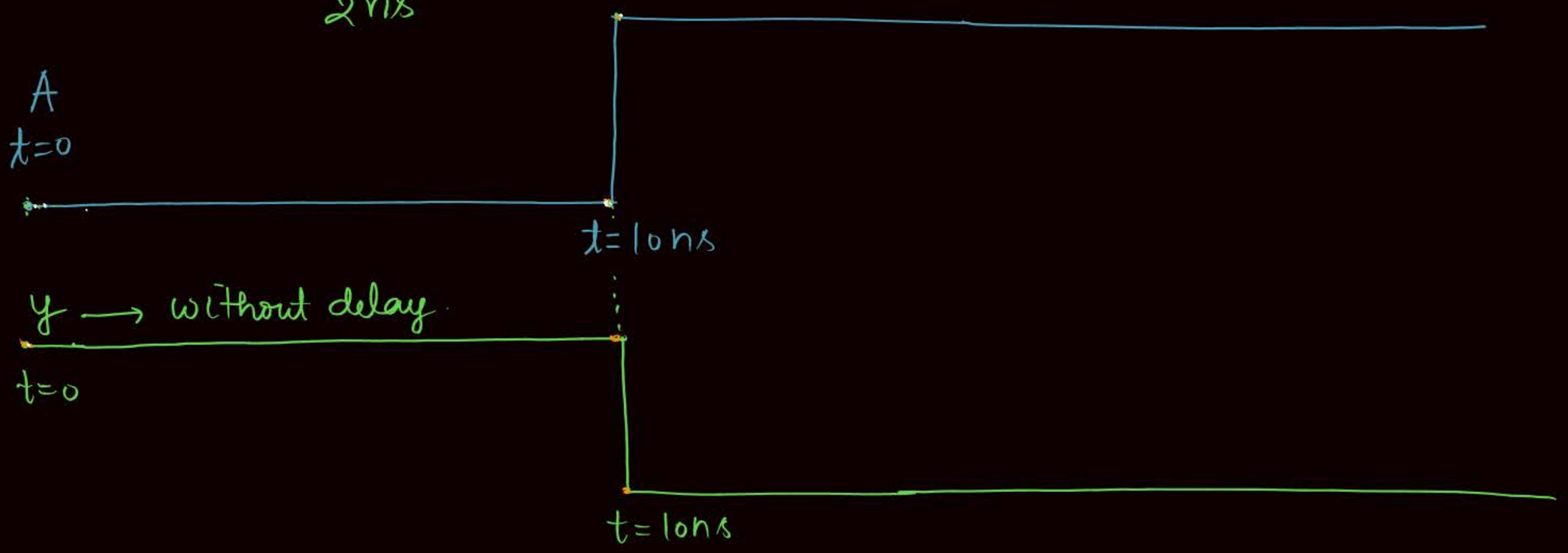
1

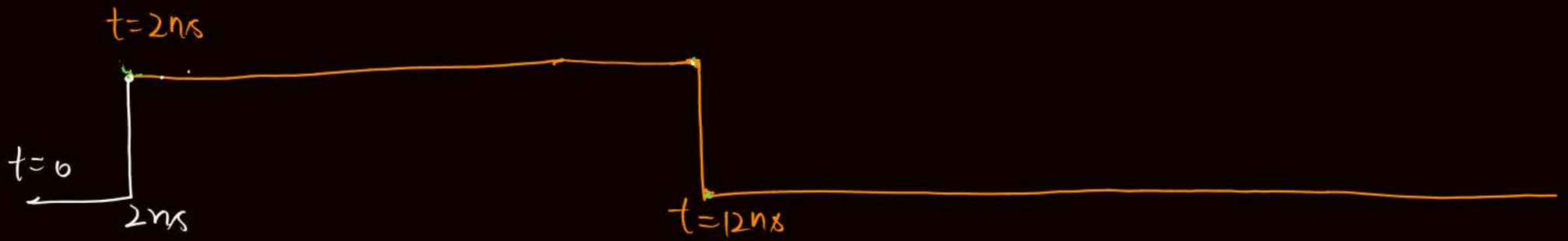
0 1 — 1 1
(1) (0)

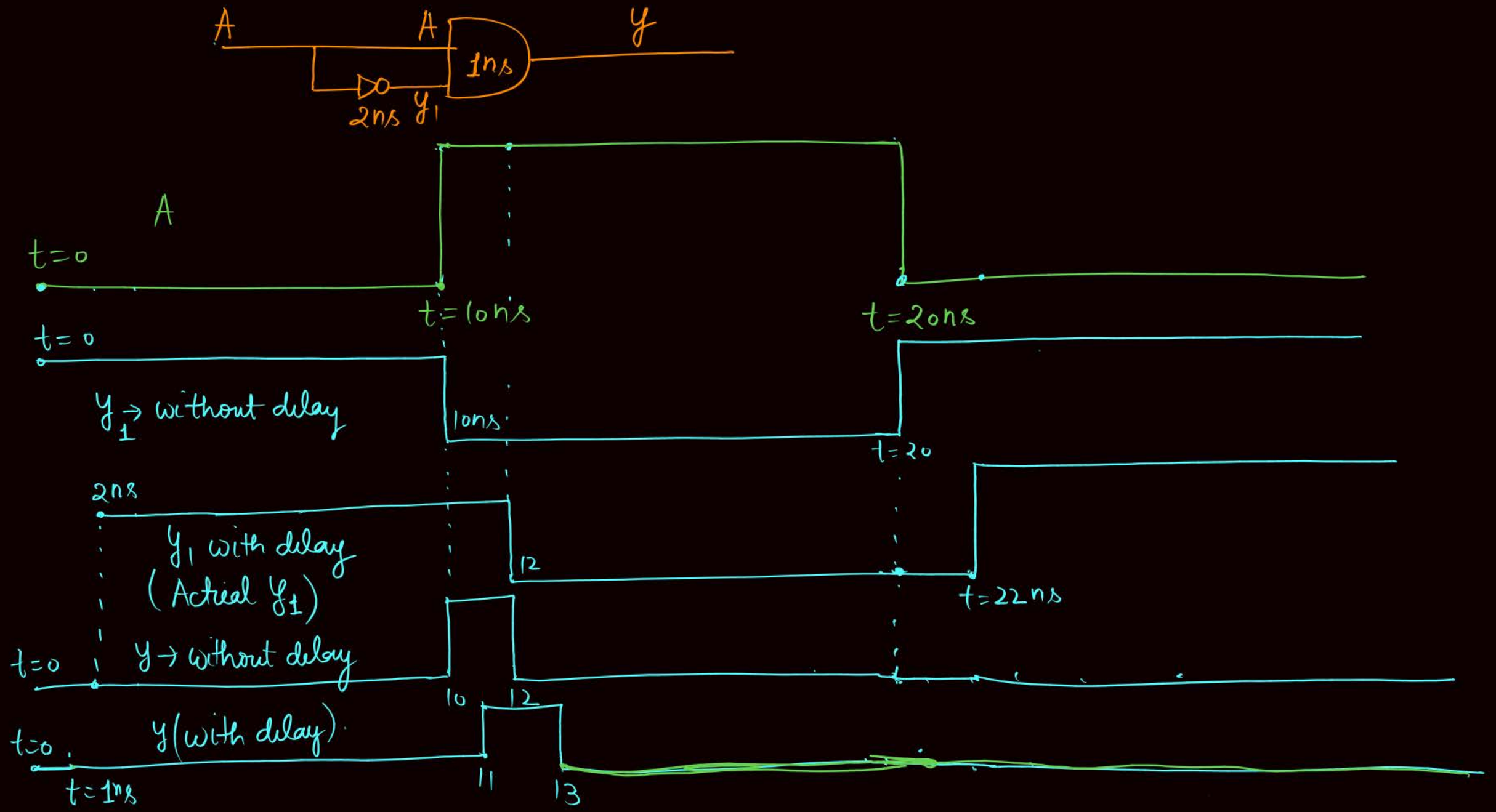
Not T_0, Not T_1, Not SD, Not L, Not M

Concept of Delay



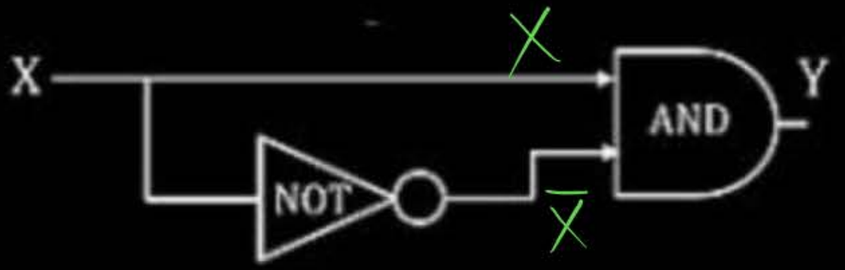





A
A
y
1ns
2ns
y_1
A
t=0
t=10ns
t=20ns
t=0
y_1 → without delay
10ns
t=20
2ns
y_1 with delay
(Actual y_1)
12
t=22ns
t=0
y → without delay
10
12
t=0
y (with delay)
t=1ns
11
13

[MSQ]

Consider the circuit shown below where the gates may have propagation delays. Assume that all signal transitions occur instantaneously and that wires have no delays. Which of the following statements about the circuit is/are CORRECT?



$X \cdot \overline{X} = 0$

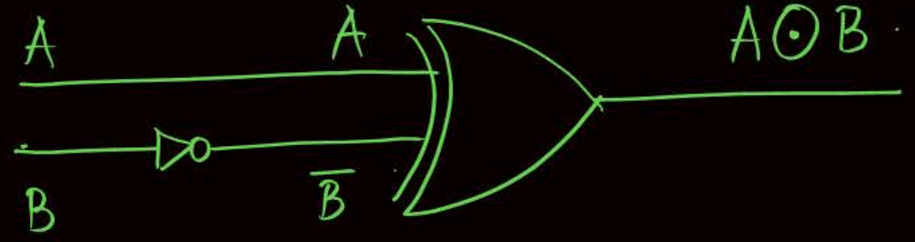
(C, D)

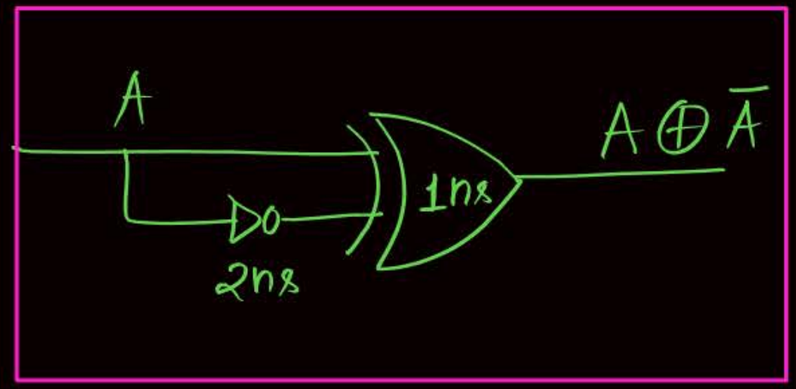
A. With propagation delays, the output Y can have a transient logic Zero after X transitions from logic One to logic Zero

B. With no propagation delays, the output Y is always logic One

C. With no propagation delays, the output Y is always logic Zero

D. With propagation delays, the output Y can have a transient logic One after X transitions from logic Zero to logic One





A = 1
B = 1
→ y = 1

A = 1
B = 0
→ y = (0)

H.W. Whether given functions are functionally complete or Not?

Q1. $f(A,B,C) = \{ f_1(A,B,C) = A + \overline{B}C \;,\quad f_2(A,B,C) = \overline{A} + \overline{B}C \}$

Q.2. $f(A,B) = \{ f_1(A,B) = \overline{A} + B \;,\quad f_2(A,B) = A\overline{B} \}$

Q.3 $f(A,B) = \{ f_1(A,B) = A\overline{B} \;,\quad f_2(A,B) = \overline{A} \}$

Q.4. $f(A,B) = \{ f_1(A,B) = (\overline{A} + B) \;,\quad f_2(A,B) = AB \}$



Topic : 2 Min Summary

- Functionally complete function
- Concept of delay.

Thank you

GW

Soldiers !

